

E02 Example Problem: CSTRs in series

The liquid phase reaction $A \rightarrow B$ is carried out isothermally and isobarically in a CSTR to 60% conversion. If a second CSTR is added downstream of the first CSTR, what should the volume of that CSTR be to achieve an overall conversion of 90%? Pure A is fed to the first CSTR in both cases.

Decision Guide: CSTRs in Series

Goal

Determine the volume of a second CSTR required to increase the overall conversion from:

60% → 90%

when placed downstream of an existing CSTR.

Step 1: Identify the Reactor Configuration

Ask:

How many reactors?

→ Two CSTRs

In parallel or in series?

→ In series

The outlet stream from Reactor 1 becomes the inlet stream to Reactor 2.

Step 2: Identify What Is Already Known

The first CSTR achieves: $X_1 = 0.60$

The desired overall conversion is: $X_{\text{overall}} = 0.90$

The feed to Reactor 1 is pure A.

Step 3: Write the CSTR Design Equation

For any CSTR:

$$V = \frac{F_{A0}(X_{\text{out}} - X_{\text{in}})}{(-r_A)_{\text{out}}}$$

CBE 3610 Reactor Design Handbook Equation (21)

This equation must be applied separately to each reactor.

Step 4: Analyze Reactor 1

For the first reactor:

$$X_{in} = 0$$

$$X_{out} = 0.60$$

Apply Equation (21).

This establishes the design equation for Reactor 1.

Step 5: Determine the Inlet Conditions to Reactor 2

A common mistake is to assume: $X_{in} = 0$ for the second reactor. This is incorrect.

The second reactor receives the outlet stream from the first reactor.

$$\text{Therefore: } X_{in,2} = 0.60$$

$$X_{out,2} = 0.90$$

Step 6: Write the Design Equation for Reactor 2

Apply the CSTR design equation again:

$$V_2 = \frac{F_{A0}(0.90 - 0.60)}{(-r_A)_{X=0.90}}$$

CBE 3610 Reactor Design Handbook Equation (21)

Notice that the rate must be evaluated at $X = 0.90$ because a CSTR always operates at outlet conditions.

Step 7: Express the Rate in Terms of Conversion

Write the rate law:

$$-r_A = [k(T)][f(C_A)]$$

using:

CBE 3610 Reactor Design Handbook Equation (44)

Then write concentration as a function of conversion.

For a liquid-phase system:

$$C_A = C_{A0}(1-X)$$

Substitute:

$$X = 0.60$$

for Reactor 1 and

$$X = 0.90$$

for Reactor 2.

Step 8: Compare the Two Design Equations

Many students expect:

$$V_2 = V_1$$

but that is usually not true.

Why?

At:

$$X = 0.90$$

the reactant concentration is much lower than at:

$$X = 0.60$$

which means:

Reaction rate is slower.

A slower rate generally requires a larger reactor volume.

Step 9: Solve for the Unknown Volume

Use:

- CSTR design equation
- Rate law
- Concentration-conversion relationship

to calculate:

$$V_2 = 1.5 \cdot V_1$$

Key Insight

The second reactor does **not** start with fresh feed. Instead, the outlet stream from the first reactor becomes the inlet stream to the second reactor. That means:

- The first CSTR increases the conversion from **0% to 60%**.
- The second CSTR receives material that is already **60% converted**.
- The second CSTR must increase the conversion from **60% to 90%**.

A common mistake is to treat the second reactor as though it were receiving fresh feed and calculate it using a conversion range of 0% to 90%.

Instead, the second reactor only achieves the **additional conversion** between 60% and 90%.

When reactors are connected in series, each reactor begins where the previous reactor ended.